1(a)(i)

3.8 / 20 = 1.9 cm

(b)

density = mass / volume = 148 / 16.6 = 89 g/cm3

4(a)

speed = distance / time = 200 / 6.4 = 30 m/s

5(b)

clockwise moments = anticlockwise moments. 200 = (2.0 x 10) + (F x 60). F = 200 /

60 = 3.33 N

11(b)

Vs/Vp = Ns/Np, so Vs = (240 / 64) x 960 = 3600 V

12(b)

137 - 56 = 81

(c)

idea of three half-lives, so 36 / 8 = 9 mg